



ST25RN300

Best-in-class NCI (NFC Controller Interface) reader for payment, access control & IoT applications



Powerful and highly sensitive NCI reader with embedded MCU and advanced features

ST25RN300 is an NCI reader with an integrated MCU, offering high RF output power and excellent receiver sensitivity to achieve the largest interaction range, even with space-constrained antenna designs. Controlled output power and the integration of many advanced features simplify compliance with EMVCo® and NFC Forum requirements.

Active load modulation, combined with automatic adjustments based on field strength, maximizes communication distance and ensures interoperability. To address the evolving requirements of access control and payment applications, designers can easily choose between different ST25RN300 product versions with or without the MIFARE Classic® feature.

KEY FEATURES AND BENEFITS

- Data exchange through NCI (NFC controller interface) over I2C interface
- Active load modulation (ALM) feature improves interoperability and compliance
- Dynamic power control (DPC) with active wave shaping (AWS) controls required for NFC Forum CR13 and beyond
- Increased noise suppression and enhanced receiver

sensitivity for easier certification

- WLCCSP package (tiny footprint)

KEY APPLICATIONS

- NFC Forum compliant NFC universal devices
- EMVCo® compliant payment devices
- Access control
- Transportation

Technical features

Supported protocols

ST25RN300 supports different mandatory protocols required for card emulation and NFC reader/writer mode:

- NFC Forum NFC-A, NFC-B (ISO14443A/B up to 848 kb/s) reader/writer functionality
- NFC-F (FeliCa™ up to 424 kb/s) reader/writer functionality
- NFC-V (ISO15693) reader/writer functionality
- MIFARE Classic® IP reader/writer functionality
- NFC-A, NFC-B and NFC-F card emulation
- ISO18092 passive initiator and target

Advanced features

ST25RN300 is a powerful NCI reader with advanced features that ensure device sensitivity, robustness, and accelerate time-to-market with an easy certification and design-in process:

- **Active wave shaping** makes it easy to set registers that limit signal overshoots and undershoots, ensuring compliance with NFC Forum and EMVCo requirements for mandatory monotonicity tests
- **Dynamic power control** ensures transmitted power levels remain safe and compliant with NFC Forum limits by automatically adjusting the output power in accordance with detuning conditions caused by different antenna sizes and distances, preventing damage to tags

- **Active load modulation** allows smaller, battery-powered NFC devices to communicate with the ST25RN300 while maintaining compatibility with existing NFC standards like ISO/IEC 14443

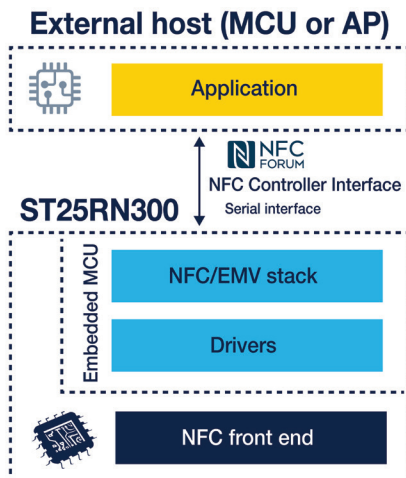
• Sophisticated tools

ST provides a sophisticated GUI, software development tools, and ready-to-use code to simplify support for embedded systems and Android processors

• Tiny package

ST25RN300 is available in a WLCSP49 (2.559mm x 2.581mm) industrial-qualified package, offering uncompromised performance in the smallest possible area for space-constrained designs.

Product architecture



Development ecosystem

Part number	Models	Interface	Serial Interface	Advance features	Output power	Ambient temperature range	Package
ST25RN300	Reader & writer, card emulation	ISO14443A/B, ISO15693, FeliCa	I2C up to 1Mbps	ALM ¹ , DPC ² , AWS ³ , NSR ⁴ , IWU ⁵	Up to 2.2 W	-40 to +85 °C	WLCSP49

Note:

1) ALM: Active load modulation, 2) DPC: Dynamic power control, 3) AWS: Active wave shaping 4) NSR: Noise suppression receiver 5) IWU: Inductive wake up.

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